multiBic® 3 mmol/L potassium solution for haemofiltration

AUSTRALIA
Package Leaflet

– Read all of this leaflet carefully before you start using multiBic® 3 mmol/L potassium.
– Keep this leaflet. You may need to read it again.
– If you have further questions please ask your doctor or pharmacist.
– If any of the side effects gets serious, or if you notice any side effects not listed in this leaflet, please tell your doctor.

In this leaflet:

1. WHAT multiBic® 3 mmol/L potassium IS AND WHAT IT IS USED FOR

multiBic® 3 mmol/L potassium is a solution for haemofiltration (removal of waste products from the body in people with kidney disease). It is used in patients with acute kidney failure. The type of solution you are given depends on the amount of potassium (a salt) in your blood. Your doctor will check your potassium levels regularly.

2. BEFORE YOU USE multiBic® 3 mmol/L potassium

Do not use multiBic® 3 mmol/L potassium

- if you have metabolic alkalosis (a condition where there is too much bicarbonate in the blood)
- if you have hypokalaemia (your potassium level is very low)
- if you are hypersensitive (allergic) to any of the ingredients, as given in Section 6.
- if haemofiltration is in general the wrong treatment in your situation for the following reasons:
  - excessive protein metabolism (hypercatabolism)
  - an impossibility to achieve a sufficient blood flow through the haemofilter
  - a high risk of bleeding related to medications required to prevent clotting in the haemofilter

Take special care with multiBic® 3 mmol/L potassium

- Your doctor will check your hydration status (the amount of water in your body), the levels of potassium, other salts, certain waste products, and your blood sugar levels. Your doctor also might advice you about your diet.
- **Do not use multiBic® 3 mmol/L potassium before mixing the two solutions in the double chamber (two-compartment) bag.**
- The tubing lines used to apply multiBic solution should be inspected every 30 minutes. If precipitate is observed within these tubing lines, bag and tubing lines must be replaced immediately and the patient carefully monitored.

Taking other medicines:

Please tell your doctor if you are taking or have recently taken any other medicines, including medicines obtained without a prescription. This **is especially important if you are taking or recently have taken digitalis (a medication prescribed for certain heart diseases).**

Pregnancy and breast-feeding

If you are pregnant or think you might be pregnant or if you are breast-feeding you should tell your doctor before starting treatment with multiBic® 3 mmol/L potassium.

3. HOW TO USE multiBic® 3 mmol/L potassium

multiBic® 3 mmol/L potassium will be given in a hospital or clinic. Your doctor will know how to use multiBic® 3 mmol/L potassium. For information intended for medical or healthcare professionals only the "Instructions for use" are given at the end of this leaflet.
For intravenous only. Sterile and free from bacterial endotoxins.

4. POSSIBLE SIDE EFFECTS

Like all medicines, multiBic® 3 mmol/L potassium can have side effects, although not everybody gets them.

The side effects of multiBic® 3 mmol/L potassium include:

- nausea (feeling sick)
- vomiting (being sick)
- muscle cramps
- changes in blood pressure

Some side effects might be caused by too much or too little fluid. These are:

- shortness of breath
- swelling of the ankles and legs
- dehydration (e.g. dizziness, muscle cramps, feeling thirsty)
- blood disorders (e.g. abnormal salt concentrations in your blood)

If you notice any of these side effects or any adverse effects not mentioned in this leaflet please inform your doctor or the nurse looking after you immediately.

5. HOW TO STORE multiBic® 3 mmol/L potassium

Your doctor knows how to store multiBic® 3 mmol/L potassium.
Store below 30°C. Do not store below 4°C.

6. FURTHER INFORMATION

What multiBic® 3 mmol/L potassium contains

Active substances:
1000 mL of the ready-to-use solution for haemofiltration contain:

	multiBic ® 3 mmol/L potassium	Unit
Sodium chloride	6.136	g
Potassium chloride	0.2237	g
Sodium hydrogen carbonate	2.940	g
Calcium chloride dihydrate	0.2205	g
Magnesium chloride hexahydrate	0.1017	g
Glucose anhydrous as Glucose monohydrate	1.000 1.100	g g

	multiBic ® 3 mmol/L potassium	Unit
Na^+	140	mmol/L
K^+	3.0	mmol/L
Ca^{2+}	1.5	mmol/L
Mg^{2+}	0.50	mmol/L
Cl^-	112	mmol/L
HCO_3^-	35	mmol/L
Glucose	5.55	mmol/L

The other ingredients are water for injections, hydrochloric acid 25 %, and carbon dioxide. multiBic® 3 mmol/L potassium is isotonic .

What multiBic® 3 mmol/L potassium looks like and contents of the pack

multiBic® 3 mmol/L potassium is delivered in a double chamber bag (two-compartment). Mixing of both solutions by opening the seam between the two chambers results in the ready-to-use solution for haemofiltration.
Each bag contains 5000 mL solution in total.

Pack size:
2 bags of 5000 mL

Marketing Authorisation Holder

Fresenius Medical Care Deutschland GmbH, 61346 Bad Homburg v.d.H., Germany

Manufacturer

Fresenius Medical Care Deutschland GmbH, Frankfurter Straße 6–8, 66606 St. Wendel, Germany

Imported and Distributed by:

Fresenius Medical Care
Australia Pty. Ltd.
305 Woodpark Road, Smithfield NSW 2164, Sydney, Australia

This leaflet was revised in December 2014.

The following information is intended for medical or healthcare professionals only:

Instructions for use

1. Removal of the overwrap and careful inspection of the bag containing the haemofiltration solution

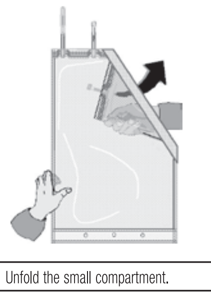
   The overwrap should only be removed immediately before using the solution.

   Plastic containers may occasionally be damaged during transport from the manufacturer to the dialysis clinic or hospital clinic or within the clinic itself. This can lead to contamination and the growth of bacteria or a fungus in the solution for haemofiltration. Therefore careful inspection of the bag and of the solution before use is necessary. Particular attention should be paid to even the slightest damage to the closure of the bag, the welded seam and the corners of the bag. The solution should only be used if clear and colourless and if the container (the bag) and connectors are undamaged and are intact.

   In case of doubt the doctor should decide whether the haemofiltration solution should be used or not.
2. Mixing of the solutions in the double chamber bag.

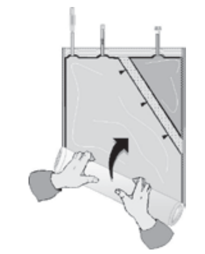
   The two solutions should be mixed immediately before use to produce a solution ready-to-use. The solution is clear and colourless.

A)



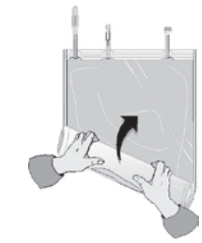
Unfold the small compartment.

B)



Roll up the solution bag starting from the corner opposite the small compartment ...

C)



… until the peel seam between both compartments has opened along its entire length and the solutions from both compartments are mixed.

After mixing both compartments, it must be checked, that the peel seam is completely open, that the solution is clear and colourless and that the container is not leaking.

3. Ready-to-use solution

   If prescribed, medicinal products (drugs) may be added to the ready to-use haemofiltration solution. This should only be carried out after the seam between the two chambers has been fully opened and the two solutions have been thoroughly mixed. The addition of any drugs to the haemofiltration solution would only be carried out on the doctor's advice and care must be taken not to introduce any contamination into the bag.

   After a drug has been added to the haemofiltration solution, the solution should be thoroughly mixed again before administration of the solution to the patient. The bags should be labeled with details of the name and amount of drug, together with the time and date.

   The ready-to-use solution shall be used immediately, not stored above +25°C and must be used within a maximum of 48 hours after mixing. Chemical and physical in-use stability of the ready-to-use solution has been demonstrated for 48 hours at 25°C. Other in-use storage times and conditions prior to use (longer than 48 hours including the duration of the treatment, higher than 25°C prior to the inlet of the pump unit) are the responsibility of the user.From a microbiological point of view, once connected to the haemofiltration circuit, and as hydrogen carbonate is present, the product shall be used immediately. Other in-use storage times and conditions are the responsibility of the user.

   The haemofiltration solution should be warmed prior to infusion with appropriate equipment to approximately body temperature and must not be infused under any circumstances below room temperature. The warming of this solution to approximately body temperature must be carefully controlled verifying that the solution is clear and without particles. If not otherwise prescribed, the ready-to-use solution for haemofiltration should be warmed immediately before infusion into the patient to 36.5°C - 38°C. The exact temperature must be selected on the doctor's advice depending on the clinical need. Only devices approved for warming haemofiltration solutions must be used. The use of a dedicated device for haemofiltration treatments which will also warm the solutions for haemofiltration prior to infusion is recommended. Microwave ovens must not be used to warm multiBic® 3 mmol/L potassium due to the possibility of local overheating.

   During application of multiBic in CRRT, white calcium carbonate precipitation has been observed in the tubing lines in rare cases, particularly close to the pump unit and the heating unit warming multiBic. Precipitations particularly can occur if the temperature of the multiBic solution at the inlet of the pump unit is already higher than 25°C. Therefore, the multiBic solution in the tubing lines should be closely visually inspected every 30 min during CRRT in order to ensure, that the solution in the tubing system is clear and free from precipitate.

   Precipitations may occur also with substantial delay after start of treatment. If precipitate is observed, multiBic solution and CRRT tubing lines must be replaced immediately and the patient carefully monitored.

   The haemofiltration solution is for single use only; any unused solution and any damaged containers should be discarded.

FRESENIUS MEDICAL CARE

multiBic® 3 mmol/L potassium solution for haemofiltration

MALAYSIA
Package Leaflet

1. Product Name
multiBic® 3 mmol/l potassium Solution for haemofiltration

2. Name and Strength of Active Ingredient (s)
Active substances:
1000 mL of the ready-to-use solution for haemofiltration contain:

	multiBic® 3 mmol/l potassium	Unit
Sodium chloride	6.136	g
Potassium chloride	0.2237	g
Sodium hydrogen carbonate	2.940	g
Calcium chloride dihydrate	0.2205	g
Magnesium chloride hexahydrate	0.1017	g
Glucose anhydrous as Glucose monohydrate	1.000 1.100	g g

	multiBic® 3 mmol/l potassium	Unit
Na^+	140	mmol/L
K^+	3.0	mmol/L
Ca^{2+}	1.5	mmol/L
Mg^{2+}	0.50	mmol/L
Cl^-	112	mmol/L
HCO_3^-	35	mmol/L
Glucose	5.55	mmol/L

The other ingredients are water for injections, hydrochloric acid 25%, and carbon dioxide.

3. Product Description
multiBic® 3 mmol/l potassium is a solution for haemofiltration. The solution is clear and colourless.
Theoretical osmolarity : 298 mosm/l pH ≈ 7.2

4. Pharmacodynamics / Pharmacokinetics

4.1 Pharmacodynamic Properties
Pharmacotherapeutic group
Group: Solution for haemofiltration
ATC code: B05Z B - Haemofiltrates

Basic principles of Haemofiltration:
During continuous haemofiltration water and solutes such as uremic toxins, electrolytes, and bicarbonate are removed from the blood by ultrafiltration. The ultrafiltrate is replaced by a substitution solution (a solution for haemofiltration), with a balanced electrolyte and buffer composition.

The ready-to-use haemofiltration solution is a bicarbonate-buffered substitution solution for intravenous administration for the treatment of acute renal failure of any origin by continuous haemofiltration.

The electrolytes Na+, K+, Mg^{2+}, Ca^{2+}, Cl^-, and bicarbonate are essential for the maintenance and correction of fluids and electrolyte homeostasis (blood volume, osmotic equilibrium, acid-base balance).

4.2 Pharmacokinetic Properties
The ready-to-use haemofiltration solution must only be administered intravenously.

The distribution of electrolytes and bicarbonate is regulated in accordance with requirements and the metabolic status and residual renal function. The active substances of the substitution solution are not metabolized except for glucose.

The elimination of water and electrolytes depends on cellular requirements, the metabolic status, the residual renal function, and on other routes of fluid losses (e.g., gut, lung, and skin).

5. Indication
multiBic® 3 mmol/l potassium is used in patients with acute kidney failure requiring continuous haemofiltration.

6. Recommended Dose
Haemofiltration in patients with acute renal failure including the prescription of substitution solutions should be under the direction of a physician with experience in this treatment.

In acute renal failure, treatment is carried out for a limited period and is discontinued when renal function is restored.

7. Mode of Administration
multiBic® 3 mmol/l potassium is exclusively indicated for intravenous use.

Infuse the ready-to-use solution into the extracorporeal circulation by means of a metering pump.

As blood serum is filtered off in haemofiltration the filtered volume, minus the necessary ultrafiltration fluid must be substituted in the form of haemofiltration solution.

The filtration rate is prescribed by the attending physician depending on the clinical status and the body weight of the patient. Unless otherwise prescribed a total filtration rate of 800 to 1400 ml/h is appropriate in adults to remove metabolic waste products depending on the metabolic status of the patient. A maximum filtration rate of 75 l per day is recommended.

There is no clinical experience on the use and dosing of this product in children.

The following information is intended for medical or healthcare professionals only.

Instructions for use

1. Removal of the overwrapping and careful inspection of the bag containing the haemofiltration solution

   The overwrap should only be removed immediately before using the solution.

   Plastic containers may occasionally be damaged during transport from the manufacturer to the dialysis clinic or within the clinic itself. This can lead to contamination and microbiological or fungal growth in the haemofiltration solution. Therefore careful visual inspection of the bag and of the solution before use is necessary. Particular attention should be paid to even the slightest damage to the closure of the bag, the welded seam and the corners of the bag.

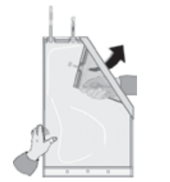
   The solution should only be used if clear and colourless and if the container and connectors are undamaged and are intact.

   In case of doubt, the doctor should decide whether the haemofiltration solution should be used or not.

2. Mixing of the solution in the double chamber bag

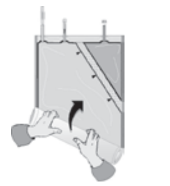
   The two solutions should be mixed immediately before use to produce a solution ready-for-use. The mixed solution is clear and colourless.

A)



Unfold the small compartment.

B)



Roll up the solution bag starting from the corner opposite the small compartment...

C)

…until the peel seam between both compartments has opened along its entire length and the solutions from both compartments are mixed.

   After mixing both compartments, it must be checked, that the peel seam is completely open, that the solution is clear and colourless and that the container is not leaking.

3. Ready-to-use solution

   If prescribed, medicinal products (drugs) may be added to the ready-to-use haemofiltration solution. This should only be carried out after the seam between the two chambers has been fully opened and the two solutions have been thoroughly mixed. The addition of any drugs to the haemofiltration solution would only be carried out on the doctor's advice and care must be taken not to introduce any contamination into the bag.

   After a drug has been added to the haemofiltration solution, the solution should be thoroughly mixed again before administration of the solution to the patient. The bags should be labeled with details of the name and amount of drug, together with the time and date.

   The ready-to-use solution should be used immediately, not stored above 25°C and must be used within a maximum of 48 hours after mixing.

   If not otherwise prescribed, the ready-to-use substitution solution should be warmed immediately before infusion to 36.5 °C – 38.0 °C. The exact temperature must be selected depending on clinical requirements and the technical equipment used.

   The haemofiltration solution is for single use; any used solution and any damaged containers should be discarded.

8. Contraindication
Solution dependent contraindication
- Hypokalaemia
- Metabolic alkalosis

Haemofiltration dependent contraindications due to the technical procedure itself:
- Renal failure with increased hypercatabolism in cases where uraemic symptoms can no longer be relieved by haemofiltration.
- Inadequate blood flow from vascular access.
- If there is a high risk of haemorrhage on account of systemic anticoagulation.

9. Warnings and Precautions
The haemofiltration solution should be warmed prior to infusion with appropriate equipment to approximately body temperature and must not be infused under any circumstances below room temperature.

The warming of this solution to approximately body temperature must be carefully controlled verifying that the solution is clear and without particles.

During application of multiBic in CRRT, white calcium carbonate precipitation has been observed in the tubing lines in rare cases, particularly close to the pump unit and the heating unit warming multiBic®. Precipitations particularly can occur if the temperature of the multiBic® solution at the inlet of the pump unit is already higher than 25°C. Therefore, the multiBic solution in the tubing lines should be closely visually inspected every 30 min during CRRT in order to ensure, that the solution in the tubing system is clear and free from precipitate.

Precipitations may occur also with substantial delay after start of treatment.

If precipitate is observed, multiBic® solution and CRRT tubing lines must be replaced immediately and the patient carefully monitored.

The serum potassium concentration must be checked regularly before and during haemofiltration. The potassium status of the patient and its trend during haemofiltration must be considered. If hypokalaemia is present or tends to develop, supplementation of potassium and / or changing to a substitution solution with higher potassium concentration may be required.

If hyperkalaemia tends to develop, an increase in the filtration rate may be indicated as well as usual measures of intensive care medicine.

In addition, the following parameters should be monitored before and during haemofiltration:

Serum sodium, serum calcium, serum magnesium serum phosphate, serum glucose, acid-base status, levels of urea and creatinine, body weight and fluid balance (for the early recognition of hyper- and dehydration).

Prior to use the solution bag must be carefully inspected as described in detail in Section 7 "Mode of Administration". Do not use before mixing the two solutions.

10.Interactions With Other Medicaments
The correct dosing of substitutions solutions and strict monitoring of clinical chemistry parameters and vital signs will avoid interactions with other drugs. The following interactions are conceivable:
- Electrolyte substitutions, parenteral nutrition and other infusions usually given in intensive care medicine interact with the serum composition and the fluid status of the patient. This must be considered when prescribing haemofiltration treatment.
- Haemofiltration treatment may reduce the blood concentration of drugs, especially of drugs with a low protein binding capacity, with a small distribution volume, with a molecular weight below the cut-off of the haemofilter and of drugs adsorbed to the haemofilter. An appropriate revision of the dose of such drugs may be required.
- Toxic effects of digitalis may be masked by hyperkalaemia, hypermagnesaemia and hypocalcaemia. The correction of these electrolytes by haemofiltration may precipitate signs and symptoms of digitalis toxicity, e.g. cardiac arrhythmia.

11.Pregnancy and Lactation
At present no clinical experience is available. The bicarbonate-buffered substitution solution must only be used after assessment of the potential risks and benefits for the mother and child.

12.Undesirable Effects
Adverse reactions, such as nausea, vomiting, muscle cramps, hypotension and hypertension, may result from the treatment mode itself or may be induced by the substitution solution.

In general, the tolerability of bicarbonate buffered haemofiltration solution is good. However, the following potential side effects of the treatment can be anticipated: Hyper- or hypohydration, electrolyte disturbances (e.g., hypokalaemia), hypophosphataemia, hyperglycaemia, and metabolic alkalosis.

13.Overdose and treatment
After use of recommended doses no reports of emergency situations have arisen; moreover, the administration of the solution can be discontinued at any time. If fluid balance is not accurately calculated and monitored, hyperhydration or dehydration may occur, with the resultant associated circulatory reactions. These may be manifest through changes in blood pressure, central venous pressure, heart rate, and pulmonary arterial pressure. In cases of hyperhydration congestive cardiac failure and/or pulmonary congestion may be induced.

In cases of hyperhydration, ultrafiltration should be increased, and the rate and volume of substitution solution infused reduced. In cases of marked dehydration, ultrafiltration should be decreased or discontinued, and the volume of substitution solution infused increased as appropriate.

Over treatment may result in disturbances of electrolyte concentrations and the acid-base-balance, e. g. an overdose of bicarbonate may occur if an inappropriate large volume of substitution solution is infused/administered. This could possibly lead to metabolic alkalosis, decrease of ionized calcium or tetany.

14.Storage Condition
Store below 30°C. Do not store below 4°C. Keep out of reach and sight of children.

Shelf life : 12 months

Chemical and physical in-use stability of the ready-to-use solution has been demonstrated for 48 hours at 25°C. Other in-use storage times and conditions prior to use (longer than 48 hours including the duration of the treatment, higher than 25°C prior to the inlet of the pump unit) are the responsibility of the user. From a microbiological point of view, once connected to the haemofiltration circuit, and as hydrogen carbonate is present, the product shall be used immediately. Other in-use storage times and conditions are the responsibility of the user

15.Dosage Forms and packaging available
multiBic® 3 mmol/l potassium is delivered in a double chamber bag (two-compartment). Mixing of both solutions by opening the seam between the two chambers result in the ready-to-use solution for haemofiltration.

Each bag contains 5000 mL solution in total.

Pack size : 2 bags of 5000 mL

16.Name and Address of Manufacturer/Marketing Authorization Holder/Product Registration Holder

Marketing Authorization Holder
Fresenius Medical Care Deutschland GmbH
61346 Bad Homburg v.d.H
Germany

Manufacturer
Fresenius Medical Care Deutschland GmbH
Frankfurter StraBe 6-8
66606 St. Wendel, Germany

Product Registration Holder
Fresenius Medical Care Malaysia Sdn. Bhd.,
2nd Floor, Axis Technology Centre,
Lot 13, Jalan 51A/225, 46100 Petaling Jaya, Selangor, Malaysia.

17.Date of Revision of Package Insert
The insert was last revised in March 2015.

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